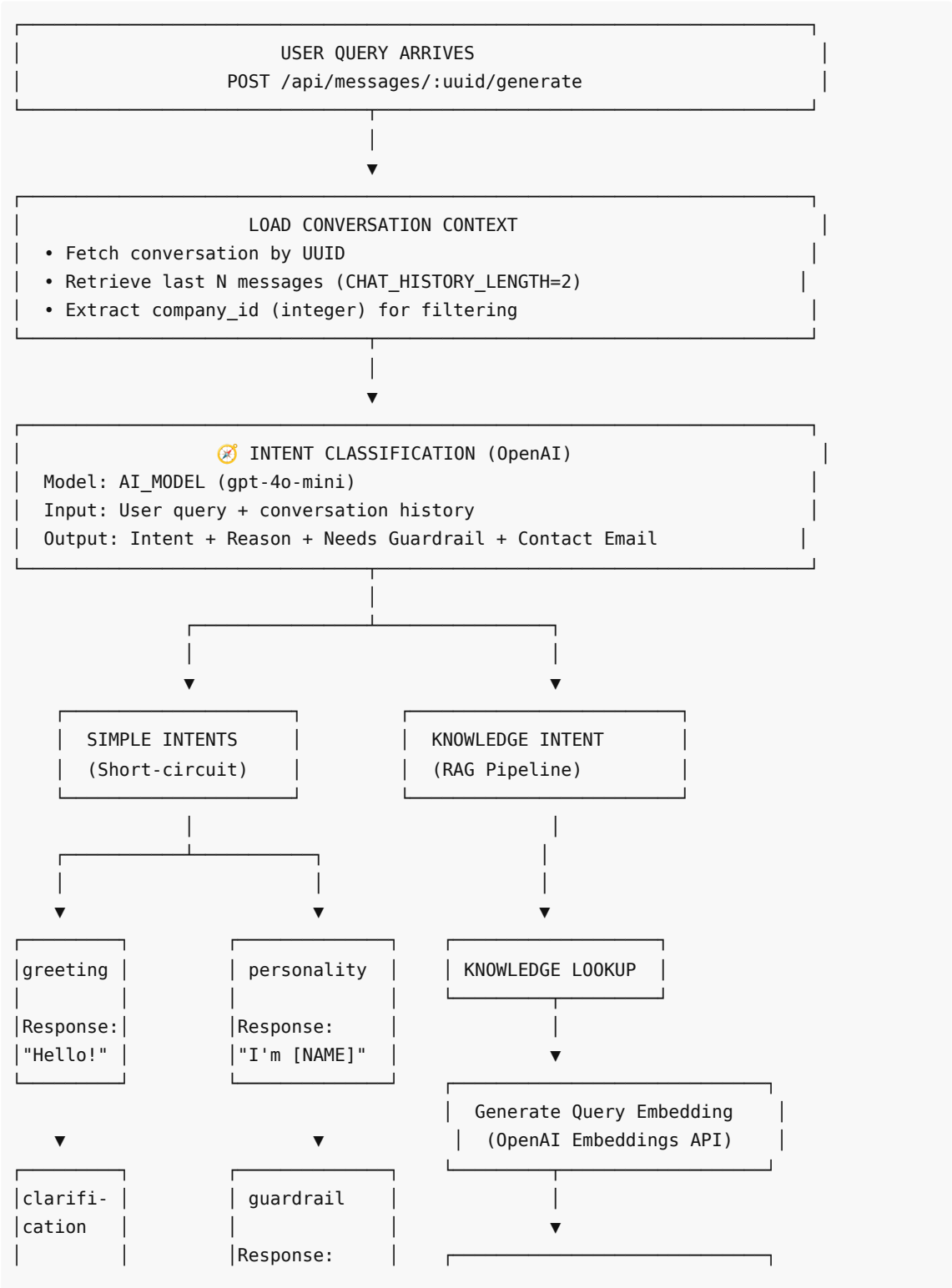
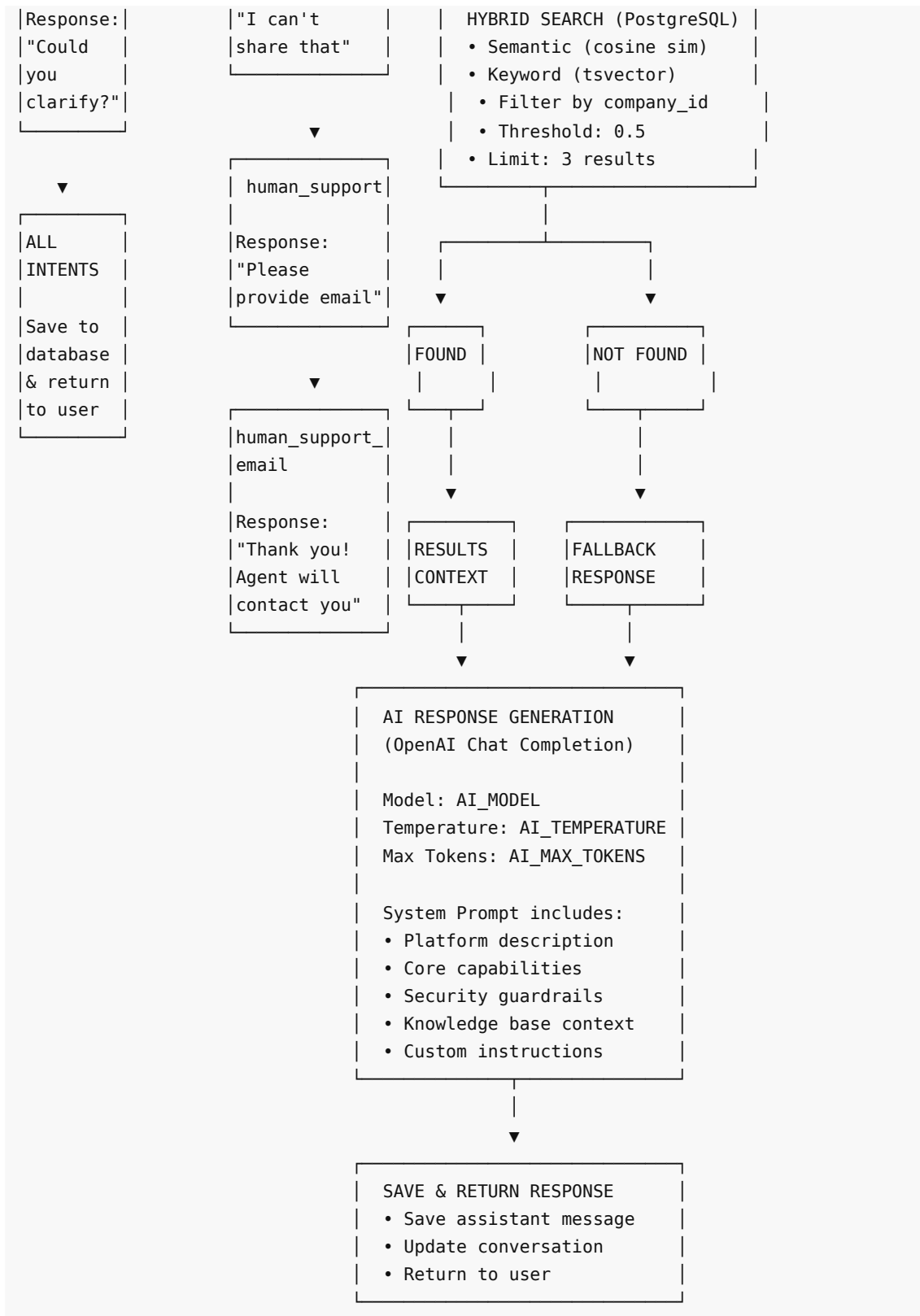


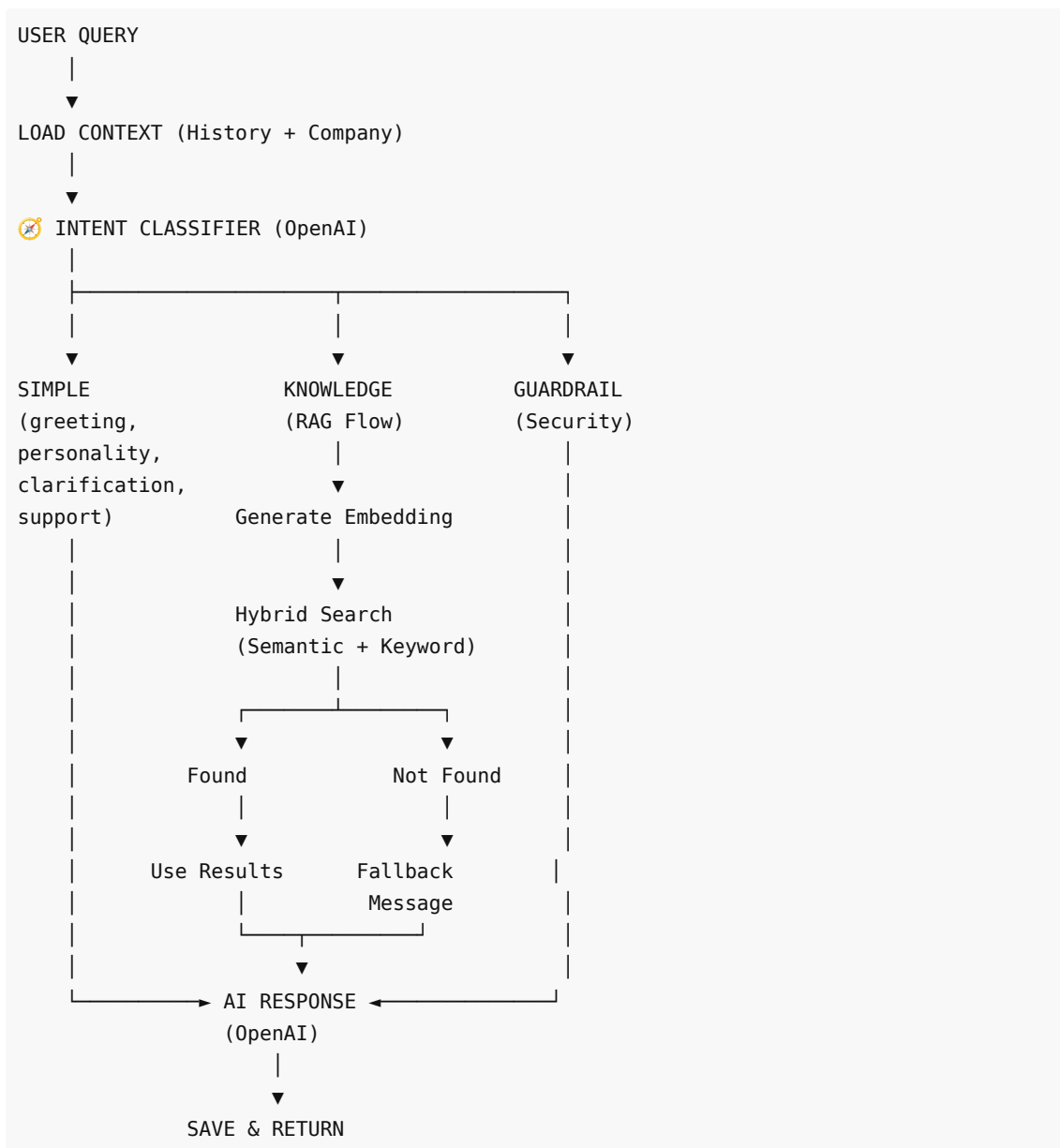
Vezlo AI Assistant - Chatbot Flow Documentation

Detailed Flow Diagram





Lightweight Flow Diagram



Key Components

1. Intent Classification

Purpose: Optimize resource usage by identifying query type before expensive RAG operations

Intents:

- **knowledge** : Questions requiring knowledge base lookup (triggers RAG)
- **greeting** : Simple greetings (direct response: "Hello!")
- **personality** : Questions about assistant identity (direct response with name)
- **clarification** : Unclear/incomplete queries (ask for clarification)
- **guardrail** : Requests for sensitive information (security refusal)
- **human_support_request** : User wants human agent (ask for email)
- **human_support_email** : User provides email (confirm handoff)

Benefits:

- Reduces OpenAI API calls for simple queries
- Prevents unnecessary database searches
- Improves response time for common interactions

2. RAG (Retrieval-Augmented Generation) Pipeline

Triggered by: `knowledge` `intent` only

Steps:

1. **Generate Embedding:** Convert query to vector (OpenAI Embeddings API)
2. **Hybrid Search:**
 - Semantic: Cosine similarity between embeddings
 - Keyword: PostgreSQL full-text search (tsvector)
 - Company Filter: Only search within user's organization
 - Threshold: 0.5 (similarity score)
 - Limit: Top 3 results
3. **Context Assembly:** Build system prompt with retrieved knowledge
4. **AI Generation:** OpenAI generates response using context

3. Security Guardrails

Implemented at two levels:

Intent Classifier:

- Detects requests for API keys, passwords, tokens, env vars
- Returns `guardrail` intent

System Prompt:

- Explicit instructions to never expose secrets
- Safe architectural guidance allowed
- Redaction of sensitive configuration

Refusal Message:

"I can help with documentation or implementation guidance, but I can't share credentials or confidential configuration. Please contact your system administrator or support for access."

4. Configuration (Environment Variables)

```
AI_MODEL=gpt-4o-mini           # Model for all OpenAI calls
AI_TEMPERATURE=0.7             # Response creativity (0-1)
AI_MAX_TOKENS=1000             # Maximum response length
CHAT_HISTORY_LENGTH=2          # Messages for context
ORGANIZATION_NAME=Your Organization
ASSISTANT_NAME=AI Assistant
PLATFORM_DESCRIPTION=...        # Custom platform description
```

5. Chat History Context

- Retrieves last N messages (default: 2)
- Provides conversation continuity

- Used by both intent classifier and response generator
- Enables multi-turn interactions (e.g., email collection)

Flow Statistics

API Calls per Query Type

Query Type	Intent Classification	Embedding Generation	AI Response	Total
Greeting	✓	-	-	1
Personality	✓	-	-	1
Clarification	✓	-	-	1
Guardrail	✓	-	-	1
Support Request	✓	-	-	1
Knowledge (found)	✓	✓	✓	3
Knowledge (not found)	✓	✓	✓	3

Performance Optimization

- **Before Intent Classifier:** Every query → 3 API calls (embed + search + generate)
- **After Intent Classifier:** Simple queries → 1 API call (60-80% reduction for typical usage)

Database Schema (Relevant Tables)

vezlo_knowledge

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `company_id` (integer): Organization filter
- `title`, `description`, `content` : Knowledge content
- `embedding` (vector): Semantic search vector
- `metadata` (jsonb): Additional data

vezlo_conversations

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `company_id` (integer): Organization filter
- `creator_id` (integer): User reference
- `title` : Conversation name

vezlo_messages

- `id` (integer): Primary key
- `uuid` (uuid): External identifier
- `conversation_id` (integer): Parent conversation
- `sender_type` : 'user' | 'assistant'
- `content` : Message text

- `parent_message_id` : Threading support

Error Handling

No Knowledge Found

When RAG search returns 0 results:

"I'm sorry, I couldn't find the requested information in my knowledge base. Please contact support for further assistance or check if the information might be available in other resources."

LLM Behavior

- System prompt instructs LLM to ONLY use knowledge base context
- If no context provided, LLM must use fallback message
- Prevents hallucination from general training knowledge

Future Improvements

1. Streaming Responses (Planned)

- Server-Sent Events (SSE)
- Real-time token streaming
- Better UX for long responses

2. Knowledge Chunks Architecture (Planned)

- New `knowledge_chunks` table
- Chunk-level embeddings
- More precise source citations
- Better relevance scoring

3. Manual Tool Calling (Discussed)

- Intent dispatcher to server-side functions
- Extensible for future capabilities (scheduling, ticketing, etc.)
- Full control over execution flow

Generated: November 11, 2025

Version: assistant-server v2.0.1

Phase: 3 (Intent Classification + RAG + Guardrails)